Third-Year Physics Laboratory

Mon-Thur Group | Experiment Descriptions | 2026–2027

Note that the list describes the starting experiments to learn about the setup. After you complete these, you are expected to come up with a new question (extension) to address using the same setup (or a modified one).

Experiment	Instructor(s)
Biophysics A & B	Nir Livne
SQUID	Einav, Rotem
Electronic Noise	Einav
Superconductivity	Einav
Spectroscopy	Rotem
DFSAS (Revaia)	Rotem
Quantum Optics	Nir Kutchuk
X-ray	Ofek
NMR	Ofek

1. Biophysics

Instructor(s): Nir Livne

Collective Motion of Bacteria

Flocks of birds, schools of fish, and swarms of insects all show striking large-scale organization: many individuals move as a single coherent unit, over length scales far larger than any one of them, and with no leader coordinating the group. What these systems share is that each individual follows a simple set of local rules—align with your neighbors, avoid collisions, stay close—and that these short-range interactions alone are enough to produce long-range order and collective behavior. Such systems are inherently out of equilibrium: every individual consumes energy in order to propel itself, so the system continuously dissipates energy and cannot be described by equilibrium statistical mechanics. As a result they exhibit phenomena with no equilibrium counterpart.

The same concept operates at the micron scale, where the local rules are not behavioral, but purely physical. Escherichia coli (E. coli) is a micron-sized bacterium that swims by rotating helical flagella. As it swims, it stirs the liquid around it, and this minute flow tends to align neighboring cells that are close enough. When the bacteria are packed densely enough that the typical distance between them becomes comparable to the range of these hydrodynamic interactions, the local alignment propagates from cell to cell and the suspension spontaneously organizes into collective motion: large correlated groups swimming together, forming swirls and jets many times larger than a single bacterium, which continuously appear, merge, and break apart. This state is sometimes called "active turbulence".

In this experiment, you will prepare dense suspensions of E. coli and record their dynamics using time-lapse microscopy. Using advanced image analysis such as Particle Image Velocimetry (PIV), you will extract the velocity field of the suspension and characterize the collective motion quantitatively.

Bacterial Swimming

Escherichia coli (E. coli) is an intestinal bacterium found in healthy humans and other animals. Because it is readily available and convenient to grow in the laboratory, it has been used to study bacteria since the beginning of modern microbiological research and is the most extensively studied bacterium to date. We now understand the activity of many cellular systems and can even measure and edit its genetic code. In many cases, conclusions drawn from research on E. coli help us learn about other bacteria and even other living organisms. A century of accumulated knowledge and genetic tools also means we can ask new and advanced questions about it.

E. coli is a micron-sized bacterium, yet it contains many systems that operate simultaneously and interact in complex ways. Among them are motors that enable the bacterium to swim and a sensory system that allows it to measure the chemical conditions in its environment and navigate accordingly. Physically, a swimming bacterium is a system out of equilibrium: it burns energy continuously to power its motors, and its motion is therefore fundamentally different from the passive, diffusive motion of a particle of the same size. This gives rise to behaviors that cannot occur in a system at equilibrium.

In this experiment, you will investigate bacterial swimming using a microscope. Since bacteria swim in three dimensions, you will reconstruct their 3D trajectories from two-dimensional movies using image analysis, and use these tracks to ask physical questions about how bacteria move, how they interact with surfaces, and how they navigate chemical gradients.

2. SQUID

Instructor(s): Einav, Rotem

In this lab, you will gain hands-on experience with a Superconducting Quantum Interference Device (SQUID) demonstration system, based on high-temperature superconductor (HTS) technology. The system operates at liquid-nitrogen temperatures, making it more accessible than helium based systems.

SQUIDs operate on the principles of quantum mechanics and superconductivity. At their core, they use a superconducting loop containing two Josephson junctions. These junctions allow superconducting electron pairs to pass through a quantum tunneling process. When exposed to an external magnetic field, the SQUID detects changes in magnetic flux through the loop and generates a voltage response. This sensitivity arises from flux quantization: the magnetic flux through the superconducting loop is quantized in discrete units, enabling SQUIDs to measure fields as weak as a few femtoteslas. The lab will introduce you to SQUID characterization, including magnetic-field sensing.

SQUIDs are widely used in geophysics, biomagnetism, and materials science. For example, they are integral to magnetoencephalography (MEG) systems, which record the magnetic activity of the brain. In materials research, SQUIDs enable precise magnetic characterization of samples and advance our understanding of magnetic phenomena in condensed-matter physics (see the Anahory lab here at HUJI for more information).

Working with the HTS SQUID system provides an opportunity to bridge theoretical knowledge and real-world applications. You will learn to calibrate the system, measure weak magnetic fields, and analyze the resulting data. The lab also emphasizes the challenges and advantages of high-temperature superconductivity.

This experience aims to deepen your understanding of superconductivity, quantum phenomena, and experimental techniques in physics, while providing a foundation for exploring the intersection of quantum mechanics and practical instrumentation.

3. Electronic Noise

Instructor(s): Einav

Electrical measurements are never perfectly quiet. Even when a resistor is connected to an ideal measurement circuit and no external signal is applied, small voltage fluctuations are always present. Some of this noise is produced by the random thermal motion of charge carriers and is therefore described by statistical physics. In this experiment, you will measure the electrical noise generated by resistors and investigate how it depends on physical parameters such as resistance, temperature, and measurement bandwidth. In particular, you will study Johnson–Nyquist noise, whose magnitude is determined by the resistor temperature and resistance. Because the signals are very small, the experiment also introduces important practical aspects of precision measurement. You will use amplification and spectral analysis to distinguish the resistor noise from background noise generated by the electronics and the environment, and examine how averaging and bandwidth affect the measured fluctuations. The experiment connects microscopic thermal motion with a directly measurable electrical signal and provides hands-on experience with concepts such as random fluctuations, noise spectra, signal-to-noise ratio, bandwidth, and the statistical analysis of weak signals.

4. Superconductivity

Instructor(s): Einav

At the beginning of the 20th century, in his laboratory in Leiden, the Netherlands, Kamerlingh Onnes was the first to liquefy helium, allowing him to cool materials to 4.2 degrees above absolute zero. While studying the effect of temperature on the electrical resistance of mercury, he discovered a striking phenomenon: at very low temperatures mercury undergoes a phase transition and its resistance drops abruptly to zero. The phenomenon became known as superconductivity and was soon discovered in many additional materials.

Superconductivity attracted great scientific interest because of both its application potential—transmitting electricity without energy loss—and the theoretical puzzle behind it. Many decades passed before the first microscopic theory of superconductivity was proposed. The field later underwent major breakthroughs, including the discovery of “high-temperature superconductors”, which raised transition temperatures to roughly 100–150 K and introduced new theoretical problems that continue to challenge scientists.

In this experiment, you will work with BSCCO/YBCO materials and measure their electrical and magnetic properties while cooling them with liquid nitrogen. You will identify the superconducting transition and may also perform measurements related to more exotic phenomena in high-temperature and type-II superconductors.

5. Optical Spectroscopy

Instructor(s): Rotem

In the quantum world, atoms and molecules can occupy only discrete energy levels. When an atom transitions between these levels, it emits or absorbs a photon whose energy is exactly equal to the energy difference between the levels. Therefore, the light emitted by an atom or molecule contains a series of spectral lines that form a unique “fingerprint” of the material.

In this experiment, we will investigate how the quantum structure of matter is reflected in the spectrum of the light it emits. We will begin with the simple case of individual atoms, where we will see how spin–orbit interaction causes fine splitting of the energy levels, which appears as splitting of spectral lines. The light sources will be discharge tubes containing different gases, including alkali elements whose effective behavior resembles that of the hydrogen atom.

We will then examine the continuous black-body spectrum from light sources such as halogen, fluorescent, and incandescent lamps, and observe how temperature affects the energy distribution of the emitted photons.

As an extension, molecular spectra can be analyzed. These reveal more complex phenomena such as fluorescence, in which part of the excitation energy is converted to heat before photon emission, as well as light absorption and changes in spectrum and intensity according to the Beer–Lambert law.

Additional extensions may include imaging spectroscopy, in which a two-dimensional image is used to study the spectral composition of its pixels, allowing maximal information to be extracted from the image and a range of properties to be investigated.

The experiment demonstrates how fundamental quantum phenomena appear in measurable properties of matter and provides experience with advanced measurement tools.

6. Probing Rubidium Atoms with Laser Light — DFSAS (Revaia)

Instructor(s): Rotem

One of the main challenges in precision physics is instrument calibration. A leading technique for calibrating lasers is to match the laser frequency to the precisely known energies of atomic transitions. In practice, however, this is not so simple. At room temperature, atoms are constantly moving, causing their transition frequencies to be Doppler shifted. The resulting Doppler broadening can be much larger than the intrinsic width of the atomic transitions, hiding the narrow spectral features that we would like to use as precise frequency references. Doppler-free saturated absorption spectroscopy is a clever technique for overcoming this limitation, allowing us to resolve narrow atomic transitions within the much broader Doppler-broadened absorption spectrum. In this experiment, you will work with an optical bench, a tunable laser, a Fabry–Pérot cavity, and a rubidium vapor cell to perform Doppler-free saturated absorption spectroscopy (DFSAS) and measure the hyperfine transitions of rubidium.

7. Quantum Optics

Instructor(s): Nir Kutchuk

Quantum optics is at the forefront of quantum technologies and also enables fundamental research into quantum theory. In this experiment, you will experimentally demonstrate fundamental elements of quantum optics and see how the theoretical material from the basic quantum-mechanics course appears in laboratory measurements.

In particular, you will reproduce a series of foundational experiments in the development of quantum optics:

a. Demonstrate the particle-like behavior of a photon using an HBT measurement.

b. Demonstrate the wave-like behavior of a single photon, and how it interferes with itself in a Michelson interferometer.

c. Demonstrate photon entanglement using a Bell inequality.

d. Demonstrate two-photon interference and explore the idea of photon indistinguishability using the Hong–Ou–Mandel effect.

8. X-ray Diffraction

Instructor(s): Ofek

X-rays are high-energy electromagnetic waves with wavelengths comparable to interatomic distances, making them an ideal tool for investigating the microscopic structure of solid matter. By observing how these waves scatter, we can deduce the precise geometric arrangement of atoms within a material.

You will explore the principles of crystallography and X-ray diffraction through three experiments:

1. Demonstrate Bragg reflection by scattering monochromatic X-rays off a single crystal at varying angles, allowing you to measure the spacing between atomic planes and verify Bragg's law.
2. Demonstrate the Laue method by directing a continuous spectrum of X-rays onto a stationary single crystal to produce a distinct pattern of diffraction spots, which reveals the internal symmetry of the crystal lattice.
3. Demonstrate the Debye-Scherrer method by exposing a polycrystalline powder sample to monochromatic X-rays to produce concentric diffraction rings, enabling you to determine the lattice structure and parameters of randomly oriented microcrystals.

9. Nuclear Magnetic Resonance (NMR)

Instructor(s): Ofek

Nuclear Magnetic Resonance (NMR) is a physical phenomenon observed when nuclei with nonzero spin are placed in a strong, constant magnetic field and perturbed by weak radiofrequency pulses. When the applied frequency matches the natural frequency of the magnetic system, it becomes possible to measure magnetization at the atomic level, providing insights into the molecular structure of matter. NMR is widely used in chemistry, biochemistry, and structural biology to determine molecular structures in solution or in the solid state, making it one of the most powerful tools for studying molecular complexity.

The experiment begins with measurements to determine the magnetization relaxation times (T1 and T2) of mineral oil and distilled water. Students then proceed to an advanced NMR application of their choice, such as one-dimensional imaging, based on the system’s capabilities.